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Palaeomagnetic and geochronological data from Late Mesoproterozoic redbed sedimentary rocks on the western margin of Kalahari craton

JENNIFER KASBOHM[1,3]*, DAVID A. D. EVANS[1], JOSEPH E. PANZIK[1], MANDY HOFMANN[2] & ULF LINNEMANN[2]

[1]*Department of Geology and Geophysics, Yale University, New Haven, CT 06520-8109, USA*

[2]*Museum of Mineralogy and Geology, Senckenberg Natural History Collections Dresden, Königsbrücker Landstrasse 159, D-01109 Dresden, Germany*

[3]*Present Address: Department of Geosciences, Princeton University, Princeton, NJ 08544, USA*

**Corresponding author (e-mail: jkasbohm@princeton.edu)*

Abstract: Redbeds of the Aubures Formation constitute the uppermost stratigraphic unit in the Mesoproterozoic Sinclair succession of southern Namibia. Aubures palaeomagnetic remanence vectors, held almost exclusively by hematite, document at least one geomagnetic polarity reversal in the stratigraphy, a positive intraformational conglomerate test indicating primary magnetization and greatest concentration of characteristic directions at 50–60% untilting, indicating that deformation was coincident with sedimentation. The new Aubures palaeomagnetic pole, at 56.4°N and 018.0°E with $A_{95} = 11.3°$, is located on the apparent polar wander path of the Kalahari craton, between poles of the 1110 Ma Umkondo igneous event and the *c.* 1090 Ma Kalkpunt redbeds of the Koras Group near Upington, South Africa. This distinctive concordance suggests that Aubures sediments have an age of approximately 1100 Ma, that the Sinclair region was probably part of Kalahari at that time and that the Aubures and Kalkpunt redbeds are broadly correlative. New laser-ablation inductively coupled plasma mass spectrometry detrital zircon results from the Aubures Formation, including a youngest age component (1108 ± 9 Ma) that is coincident with the Kalahari-wide Umkondo large igneous province, conform to this interpretation. Palaeomagnetism and geochronology of the Sinclair succession can provide kinematic constraints on the tectonic evolution of Kalahari as it approached other cratons in the growing Rodinia supercontinent.

The Kalahari craton, in southern Africa, is thought to have played an important role in the early Neoproterozoic Rodinia supercontinent reconstructions (Hoffman 1991), because it was affected by widespread *c.* 1.2–1.0 Ga events of the Namaqua–Natal orogen (Thomas *et al.* 1994) followed by *c.* 0.8–0.7 Ga rifting (Frimmel *et al.* 1996; Macdonald *et al.* 2010). The particular location and orientation of Kalahari within the Rodinia assemblage, however, has been debated (Hanson *et al.* 2004; Jacobs *et al.* 2008; Loewy *et al.* 2011; Li *et al.* 2013). Palaeomagnetism of Proterozoic rocks can provide important insights into global and regional continental reconstructions, as well as the behaviour of the ancient geomagnetic field. Along the western margin of the Kalahari craton in southern Namibia (Fig. 1), late Mesoproterozoic volcanic and sedimentary rocks of the Sinclair region are regionally exposed and preserved at subgreenschist metamorphic grade (Miller 2008), and thus amenable to palaeomagnetic study. Pioneering palaeomagnetic work in the Sinclair succession (Piper 1975) was limited to analysis of either the natural remanent magnetization (NRM) of the rocks, or limited blanket AF demagnetization to 1600 Oe. The present report is part of a broad-scale palaeomagnetic study of the Sinclair region, using modern methods of vector analysis and field stability tests, beginning with the youngest stratigraphic unit, the Aubures Formation. In addition, we supplement the palaeomagnetic work with detrital-zircon provenance analysis for constraints on sedimentary source regions and a maximum depositional age for that redbed unit.

Geological setting

The western Mesoproterozoic margin of Kalahari craton, known as the Namaqua Province, contains several distinct terranes bounded by NW–SE dextral shear zones extending from central Namibia into western South Africa (Miller 2012). The present study area lies within the low-grade Konkiep Subprovince, located in south-central Namibia, between the largely buried Namaqua tectonic front to the NE and the Excelsior–Lord Hill Lineament

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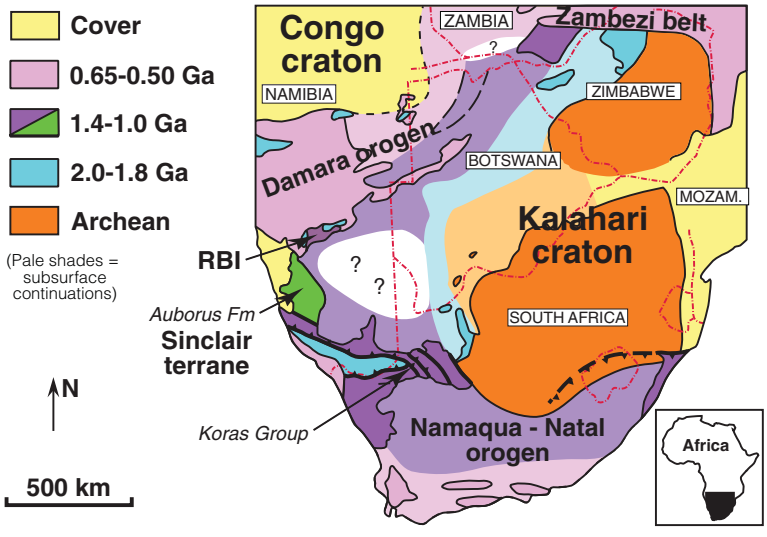


Fig. 1. Simplified tectonic map of southern Africa, modified from Hanson (2003). RBI, Rehoboth basement inlier.

to the SW. The Namaqua Front has been interpreted as a *c.* 1.4 Ga suture zone between the Konkiep Subprovince and the Kalahari craton (Miller 2008; 2012), although subsequent dextral offset of its continuation into the Upington region of South Africa has been estimated at about 200 km (Stowe 1983), possibly related to transform motion coincident with collision between Kalahari and other cratons during Rodinia assembly (Jacobs *et al.* 1993, 2008).

Within the Konkiep Subprovince, the Sinclair mining district is the type locality of the Sinclair Supergroup: a succession of Mesoproterozoic volcanic and sedimentary rocks dating from *c.* 1.4 to 1.1 Ga (Watters 1977; SACS 1980; Becker *et al.* 2006; Miller 2008, 2012). This stratigraphic unit continues northward around the western Kalahari cratonic promontory into the Rehoboth basement inlier at the southern margin of the Damara orogen (Becker *et al.* 2005, 2006). Recent U–Pb ages from metavolcanic and metasedimentary rocks of the Rehoboth region (Becker & Schalk 2008; Van Schijndel *et al.* 2011) have refined correlations to the less deformed and less metamorphosed strata of the Konkiep Subprovince, and conform to the model of the latter region's accretion to Kalahari as early as *c.* 1.4 Ga, as noted above. The Konkiep Group represents the entirety of the Sinclair Supergroup in the Konkiep Subprovince. Dominated by bimodal magmatism and epicontinental sedimentation (Becker *et al.* 2006), the Konkiep Group's tectonic setting has been compared with either a rift (Borg 1988) or the overriding plate of a subduction zone (Hoal 1993; Miller 2008, 2012), but the long duration of its stratigraphic accumulation probably implies a diverse range of tectonic settings. Aside from the oldest, Kairab Formation, which occupies a transitional stratigraphic position between basement and cover (Miller 2008, 2012), subsequent formations of the Konkiep Group are only mildly deformed, and greenschist-grade metamorphism is restricted to the western Awasib Mountain Terrain, which has been interpreted as a continental margin (Hoal 1993).

Five stratigraphic sequences have been recognized in the Konkiep Group (summarized by Miller 2008, 2012). The oldest, Kairab Formation, consists of bimodal metavolcanic rocks that are considered to represent island-arc magmatism prior to accretion of the Konkiep Subprovince. The remainder of the Konkiep Group comprises several cycles of siliciclastic and volcanogenic deposits, intruded by successive generations of intermediate to bimodal plutonic rocks. In the study area, these cycles are represented by mafic volcanic and volcaniclastic rocks of the Barby Formation, which is unconformably overlain by mainly medium- to coarse-grained siliciclastic sedimentary and volcanic rocks of the Guperas Formation. The Guperas Formation is in turn unconformably overlain by the Aubures Formation redbeds. The Barby Formation has been correlated with lavas in the western, Awasib Mountains, which are intruded by plutonic rocks dated by U–Pb on zircon at 1215–1220 Ma (Hoal & Heaman 1995).

The Aubures Formation is a >2600 m-thick redbed unit deposited in north–south-oriented synclinal basins, each about 40 km in length and 10 km in width (Miller 1969, 2008, 2012; Watters 1977). The western Aubures exposure forms a prominent erosional escarpment named the Rooirand ('Red

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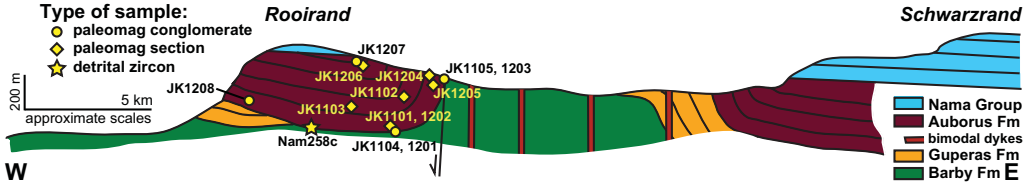


Fig. 2. Simplified and schematic cross-section of the Sinclair region.

Ridge'). The two Aubures outcrop areas are separated laterally by a *c.* 15 km-wide horst block, now exposed in a topographic low, containing lavas and sedimentary rocks from the underlying Barby and Guperas Formations (Fig. 2). The Aubures sediments were deposited in a shallow basin in which subsidence was contemporaneous with deposition, as evidenced by thick, coarse conglomerate facies localized to the Aubures-bounding faults adjacent to the central horst. The block faulting, concomitant with gentle folding, was entirely completed prior to deposition of the Ediacaran–Cambrian Nama Group, which once unconformably covered the entire Sinclair region. These successions are now discontinuously exposed as erosional remnants at the crest of the Rooirand, coplanar with continuous Nama exposures along the Schwarzrand ('Black Ridge') to the east (Fig. 2).

Physical sedimentary characteristics of the Aubures Formation indicate a fault-bounded alluvial–fluvial setting. The basal conglomerate facies, ranging in thickness from 360 to 1000 m (Miller 1969), is clast-supported, with subangular to subrounded clasts ranging in size from pebble to cobble. Most clasts are sourced from Barby lavas or rhyolite porphyries. Above the basal conglomerate lie arkosic, well-sorted sandstones and siltstones of varying thicknesses; grit beds are commonly interspersed. Bedding ranges in character from laminar units of decimetre order in thickness, indicating frequent inundations by shallow water, to massive beds up to 3 m thick. Locally, ripple-bedded shales are overlain by mudcracks, and rarely contain rain imprints, indicating intermittent exposure. Cross-bedding, ranging in thickness from 0.6 to 2 m, is also found throughout the formation; palaeocurrent directions and conglomerate clast compositions indicate multiple local sediment sources into the basins (Miller 1969).

Methods

Palaeomagnetism

More than 300 oriented blocks and individually oriented cores were collected through most of the thickness of the Aubures Formation to attempt a preliminary magnetostratigraphy of the succession. Blocks were oriented by magnetic compass, and cores were oriented by both magnetic and solar compasses. No significant deviation of magnetic declination from the regional average (−15 to −16°) was observed in any of the stratigraphic sections sampled.

The localities sampled in our palaeomagnetic study, all located within the Rooirand exposure area, are indicated in Figure 3. We sampled stratigraphic sections in different limbs and near the hinge of a broad, upright syncline in order to perform a fold test. Stratigraphic heights were logged for each palaeomagnetic sample in all sections. We also sampled both clasts and matrix from four conglomerates in the unit in order to apply conglomerate tests on the age palaeomagnetic remanence (Graham 1949). Although the majority of the sampled clasts are derived from pre-Aubures units – and thus, strictly speaking, the test applies to the age of remanence in those underlying formations – several clasts had similar lithologies to those of the Aubures redbeds and provided a more definitive comparison of palaeomagnetic remanence directions.

Samples were trimmed to *c.* 10 cm^3 specimens and analysed in the Yale Palaeomagnetic Facility using a cryogenic DC-SQuID magnetometer (sensitivity with sample holder *c.* 10^{-8} emu) with automated sample changer (Kirschvink *et al.* 2008). NRM measurements were followed by a low-temperature demagnetization step and about 20 successive high-temperature demagnetization steps (with controlled N_2 atmosphere) of decreasing intervals up to *c.* 690 °C, whereupon measured directions scattered about the origin at low intensity. Each specimen's magnetic components were resolved with principal component analysis (Kirschvink 1980) using software created by Jones (2002). Locality means, precision parameters and 95% confidence limits were calculated using Fisher (1953) statistics, inverting polarity as necessary to obtain unimodal distributions.

Detrital zircon geochronology

Zircon concentrates were separated from 2–4 kg sample material at the Senckenberg Naturhistorische

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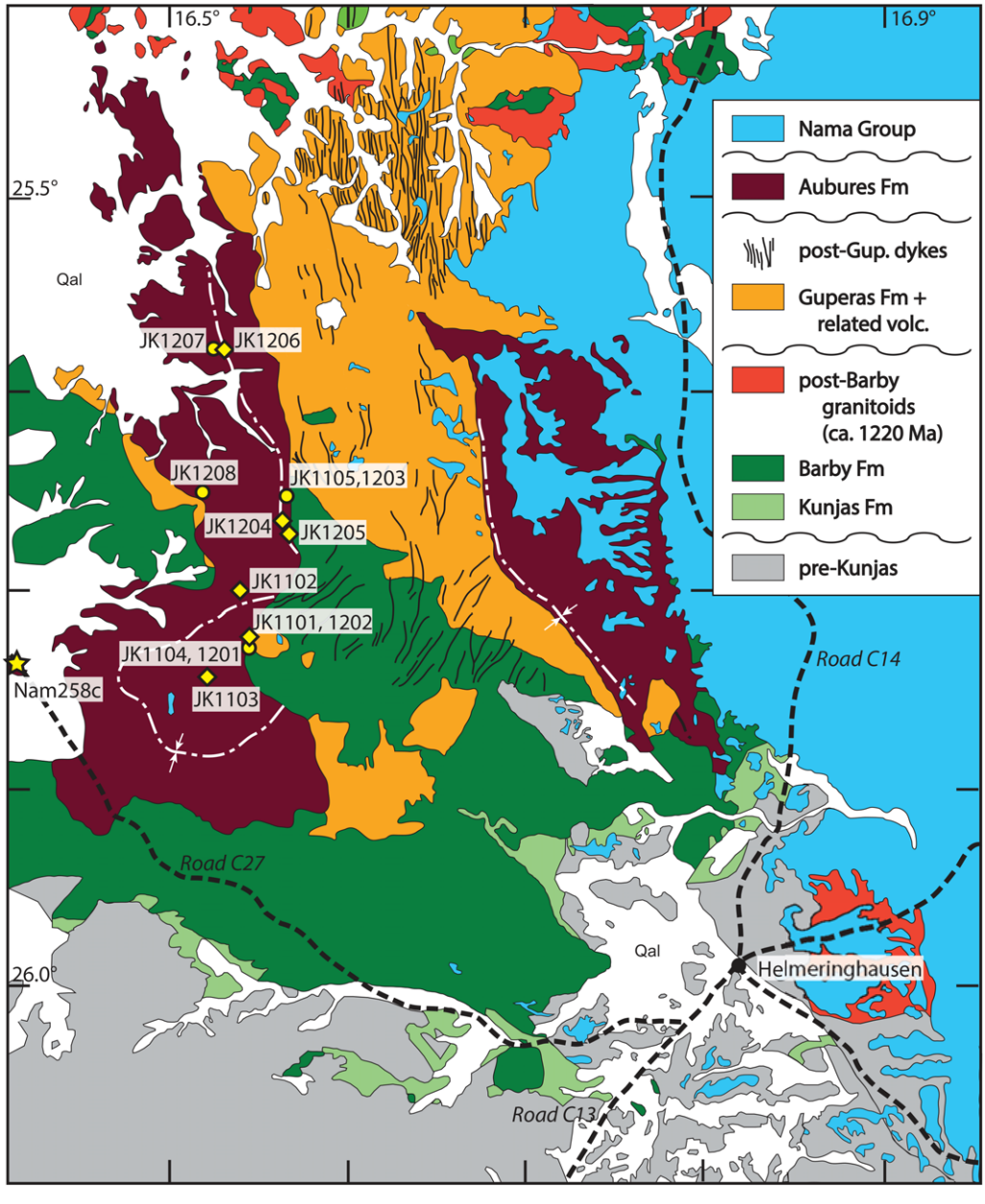


Fig. 3. Geological map of the Sinclair region, south-central Namibia. Sampling sites are denoted by symbols as shown in Figure 2. Dashed white line denotes synclinal axis. Qal, Quaternary alluvium.

Sammlungen Dresden (Museum für Mineralogie und Geologie) using standard methods. Final selection of the zircon grains for U–Pb dating was achieved by hand-picking under a binocular microscope. Zircon grains of all grain sizes and morphological types were selected, mounted in resin blocks and polished to half their thickness. Concerning stratigraphic ages the stratigraphic time scale of Gradstein *et al.* (2012) was used.

Laser ablation sector field inductively coupled plasma mass spectrometry (LA-SF-ICP-MS) techniques at the Museum für Mineralogie und Geologie

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(GeoPlasma Laboratory, Senckenberg Naturhistorische Sammlungen Dresden), using a Thermo-Scientific Element 2 XR sector field ICP-MS coupled to a New Wave UP-193 Excimer Laser System. A teardrop-shaped, low-volume laser cell constructed by Ben Jähne (Dresden) and Axel Gerdes (Frankfurt/M.) was used to enable sequential sampling of heterogeneous grains (e.g. growth zones) during time-resolved data acquisition. Each analysis consisted of 15 s background acquisition followed by 30 s data acquisition, using a laser spot size of 25 and 35 μm, respectively. A common-Pb correction based on the interference- and background-corrected ^{204}Pb signal and a model Pb composition (Stacey & Kramers 1975) was carried out if necessary. Raw data were corrected for background signal, common Pb, laser-induced elemental fractionation, instrumental mass discrimination and time-dependant elemental fractionation of Pb/Th and Pb/U using an Excel® spreadsheet program developed by Axel Gerdes (Institute of Geosciences, Johann Wolfgang Goethe-University Frankfurt, Frankfurt am Main, Germany). Reported uncertainties were propagated by quadratic addition of the external reproducibility obtained from the standard zircon GJ-1 (*c.* 0.6 and 0.5–1% for the $^{207}Pb/^{206}Pb$ and $^{206}Pb/^{238}U$, respectively) during individual analytical sessions and the within-run precision of each analysis. Concordia diagrams (2σ error ellipses) and concordia ages (95% confidence level) were produced using Isoplot/Ex 2.49 (Ludwig 2001) and frequency and relative probability plots using AgeDisplay (Sircombe 2004). The $^{207}Pb/^{206}Pb$ age was taken for interpretation for all zircons >1.0 Ga. For further details on analytical protocol and data processing see Gerdes & Zeh (2006) and Frei & Gerdes (2009). Zircons showing a degree of concordance in the range of 90–110% in this paper are classified as concordant because of the overlap of the error ellipse with the concordia. Th/U ratios are obtained from the LA-ICP-MS measurements of investigated zircon grains. U and Pb content and Th/U ratio were calculated relative to the GJ-1 zircon standard and are accurate to approximately 10%.

Results

Palaeomagnetism

Samples from almost all of the Aubures redbeds showed excellent stability to thermal demagnetization, usually with single-component behaviour and narrow unblocking temperatures (Fig. 4), with good clustering of the characteristic remanent magnetization (ChRM) directions from each locality (Table 1, Fig. 5). Our results show broad agreement with those reported by Piper (1975), but his slightly steeper mean direction of the NRM probably resulted from contamination by a minor, unresolved overprint of the recent geomagnetic field. In our study, only at locality JK1206 was there a significant low-stability component of remanence, which was characterized by a scatter of moderately steep upward directions around that of the present local field. Apart from the dominantly N-directed ChRM polarity documented at locality JK1101/JK1202 (stratigraphically lowest), the Aubures sections have mainly south-directed, shallow ChRMs. Although our sampling did not span the Aubures Formation in its entirety, the dominance of a single magnetic polarity suggests that the majority of Aubures sedimentation occurred during a time of infrequent geomagnetic reversals.

The wide scatter of characteristic directions for clasts sampled from the four conglomerate localities (Fig. 6) indicates positive conglomerate tests, whereby magnetization can be inferred to pre-date deposition of the conglomerates. The majority of sampled clasts are igneous, typically felsic porphyries that are probably derived from the Guperas Formation or pre- or post-Guperas dykes. Two clasts from one of the tests were double-sampled in the field, and showed internally consistent ChRM directions. Six clasts are reddish sandstones that are considered to have bulk lithological properties (e.g. porosity, magnetic mineralogy) most similar to the Aubures Formation redbeds, although it is uncertain whether they derive from older Aubures horizons or sedimentary rocks of the Guperas Formation; regardless, their ChRM directions widely deviate from those of the Aubures sections. The conglomerates are geographically distributed among the sampled sections and should have experienced the same post-depositional thermochemical history. For these reasons, we consider the magnetization of the Aubures redbeds to be primary for the purposes of tectonic reconstructions.

Bedding attitudes within each locality vary only slightly, so a fold test is only possible using the combination of locality means (Table 1). Partial unfolding of the Aubures dataset is illustrated in Figure 7. Although overall precision is slightly greater at 100% tilt-corrected relative to the uncorrected data, differences in *k*-values for those two coordinate systems are statistically insignificant. There is a pronounced peak in precision at 50–60% unfolding, but given the small number of sites employed in calculating this mean, the clustering is not significantly distinct at 95% uncertainty (Fisher *et al.* 1987, p. 219). Because of evidence for deposition during basinal development as noted above, and constrained further by the positive intraformational conglomerate tests, we consider the Aubures

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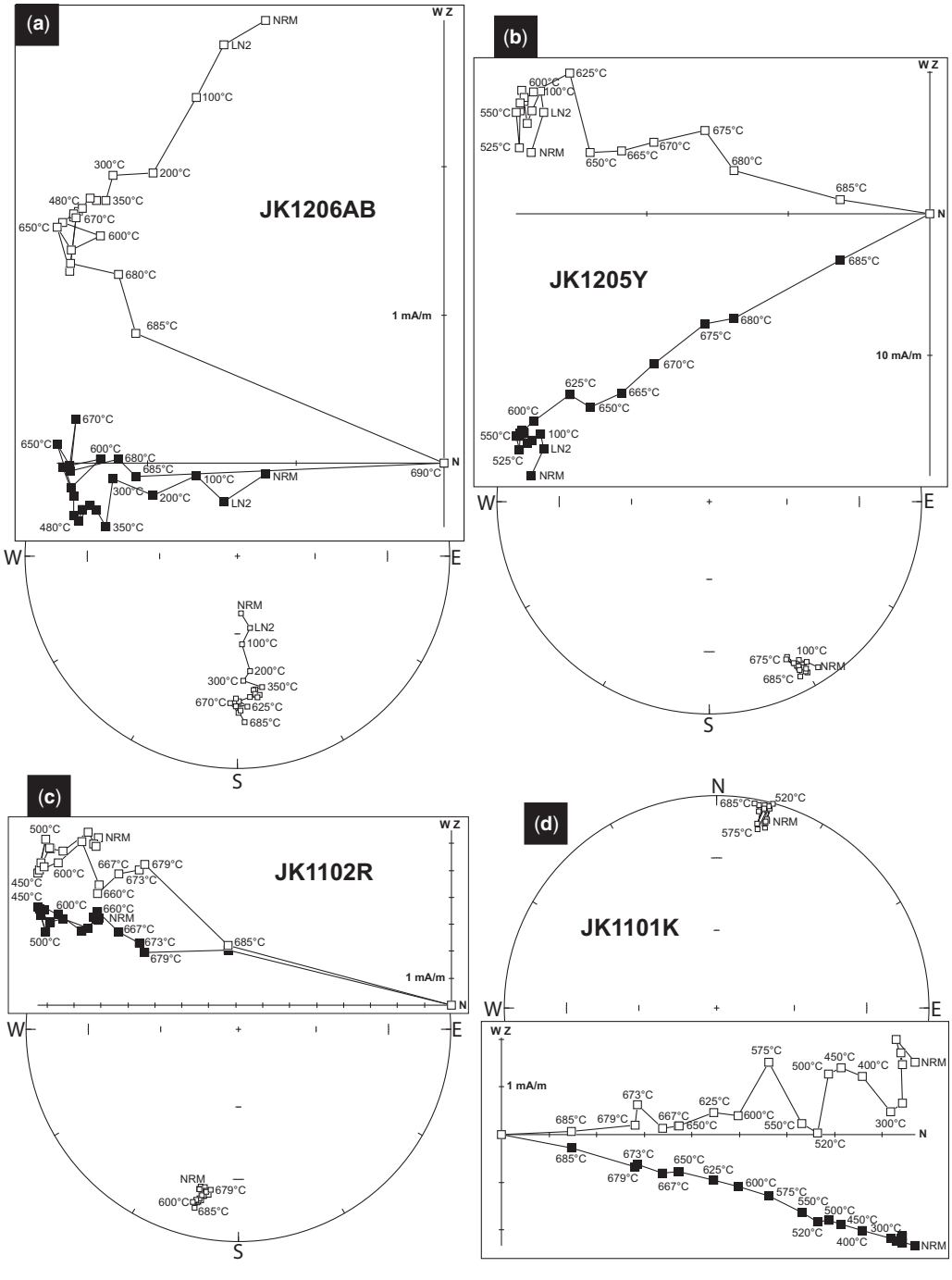


Fig. 4. Typical demagnetization behaviour for Aubures Formation redbeds. Panels are arranged in approximate geographic position. Each panel contains an orthogonal projection diagram, with solid v. open symbols representing the horizontal v. vertical plane, respectively. All data are shown in geographic coordinates. NRM, Natural remanent magnetization. LN2, Measurement following immersion in liquid nitrogen.

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Table 1. *Summary of palaeomagnetic data from the Aubures Formation, Sinclair region, Namibia*

Locality			Average bedding				Present coordinates				100% Tilt-corrected				55% Tilt-corrected					
Code	Latitude (°S)	Longitude (°E)	RHS (deg)	Dip (deg)	Strat (m)	n/N	D (deg)	I (deg)	k	α_{95} (deg)	D (deg)	I (deg)	k	α_{95} (deg)	D (deg)	I (deg)	k	α_{95} (deg)	VGP latitude (°N)	VGP longitude (°E)
JK1101, JK1202	25.73	16.55	207	34	36	23/23	010.8	02.2	37.8	5.0	015.3	−16.9	31.2	5.5	011.8	−08.6	34.8	5.2	65.8	046.4
JK1102	25.70	16.54	067	44	130	33/34	186.9	22.7	15.3	6.6	186.5	−19.4	17.7	6.1	185.8	−00.4	17.4	6.2	63.5	029.6
JK1103	25.74	16.52	197	15	128	11/13	178.2	−19.2	33.9	8.0	175.9	−11.2	28.1	8.8	176.7	−14.9	31.1	8.3	56.5	010.6
JK1204	25.67	16.56	130	38	190	32/32	169.0	−16.1	13.5	7.2	156.2	−27.6	14.3	7.0	163.0	−23.4	16.0	6.6	48.7	350.9
JK1205	25.67	16.56	171	103	202	48/49	194.7	−08.0	15.2	5.5	157.6	−20.8	15.0	5.5	177.8	−23.9	15.1	5.5	51.8	013.1
JK1206	25.58	16.53	254	12	171	71/71	190.7	−40.4	15.7	4.4	189.0	−29.2	15.3	4.5	189.6	−34.3	15.7	4.4	44.6	029.3
Mean							005.0	10.7	12.8	19.5	356.9	16.0	12.5	19.7	000.8	15.0	18.4	16.0	56.4	018.0
																			K = 36.2	$A_{95} = 11.3°$

Notes: RHS, right-hand strike; Strat, thickness of sampled section; n/N, number of specimens included in the mean/number of specimens analysed; D, mean declination; I, mean inclination; k, K, Fisher's (1953) precision parameter; α_{95}, A_{95}, radius of the 95% confidence cone; VGP, virtual geomagnetic pole (inverted to Northern Hemisphere as needed).

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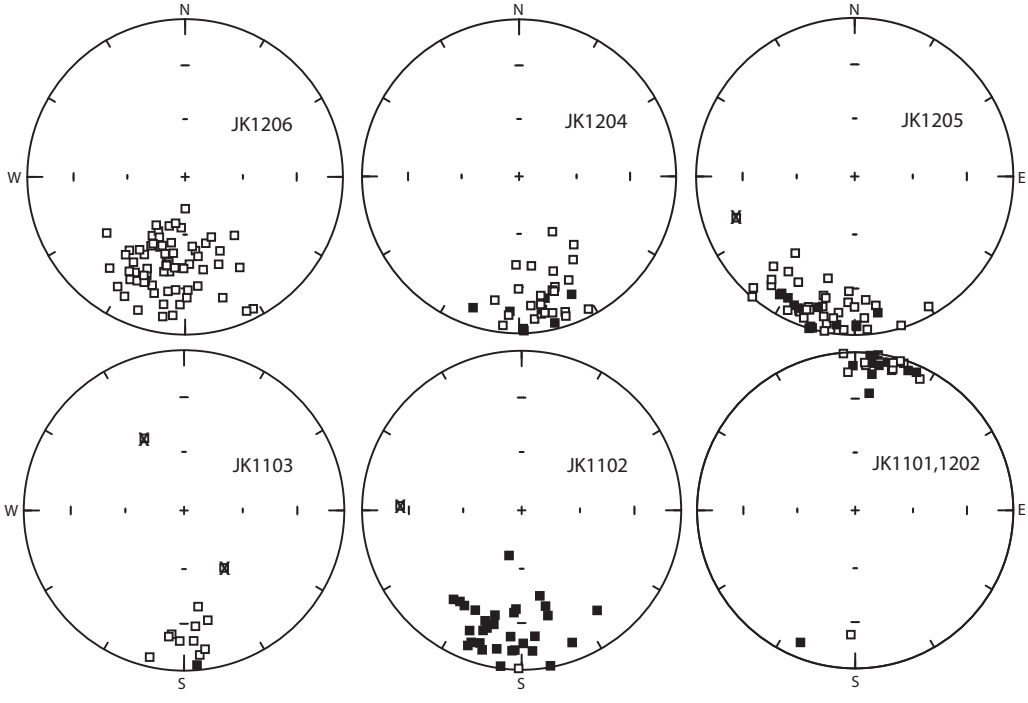


Fig. 5. Characteristic remanent magnetization directions, plotted on equal-area projections, from each of the Aubures section localities; all data are in geographic (0% unfolded) coordinates. Solid v. open symbols represent data on the lower v. upper hemisphere, respectively. Each 'X' marks an outlying datum that has been excluded from the mean.

palaeomagnetic remanence to be acquired during the medial stages of syndepositional tilting in the rift basins. Despite the statistical insignificance of the fold test, greatest concentration of directional data at 55% unfolding (Fig. 7) impels us to use that coordinate reference frame for our preferred mean direction. We note that this choice is inconsequential to our tectonic interpretations, as the 0, 55 and 100% site-mean directions are separated only by steps of about 5° (Table 1). The magnetization may be either a detrital remanence or an early diagenetic, chemical remanence, but in either case the data should be comparable to most palaeomagnetic results from ancient redbeds.

Using the peak directional clustering obtained upon 55% untilting of all localities, we calculate a palaeomagnetic pole at 56.4°N, 018.0°E, with statistical precision $K = 36.2$, and $A_{95} = 11.3°$ (Table 1). The implied palaeolatitude of deposition (or early diagenesis) is 08°. Our new Aubures palaeomagnetic pole largely confirms the result of Piper (1975), but raises its quality status considerably. On the Van der Voo (1990) seven-point quality scale, Piper's (1975) data satisfy only two criteria: statistics of the pole (#2) and structural control (#5). Our new result adds vector analysis (#3), positive field stability tests (#4) and dual polarity of the remanence (#6). The only criteria not yet satisfied are: a direct chronometric determination of the rock age (#1), although as shown below, there is compelling reason to suspect an age of *c.* 1100 Ma for Aubures sedimentation and magnetic remanence acquisition; and similarity to younger poles (#7), in particular the Middle Cambrian poles from Gondwana (Mitchell *et al.* 2010). Our positive field stability tests, however, render failure of that last criterion irrelevant to the reliability of the Aubures palaeomagnetic pole.

Detrital zircon geochronology

Detrital zircons from the Aubures Formation yielded many concordant U–Pb data (101 out of 120 analysed), with several modes of ages ranging from 2.6 to 1.1 Ga (Fig. 8, Table 2). The largest mode, taken as a broadly distributed group, spans the 1.4–1.1 Ga interval, readily explicable in terms of the active Rehoboth and Namaqua belt margins of western Kalahari (Becker *et al.* 2006; Eglington 2006; Cornell *et al.* 2012; Miller 2012). The next mode contains just nine analyses, centred on 1.9 Ga. There are scattered analyses in the

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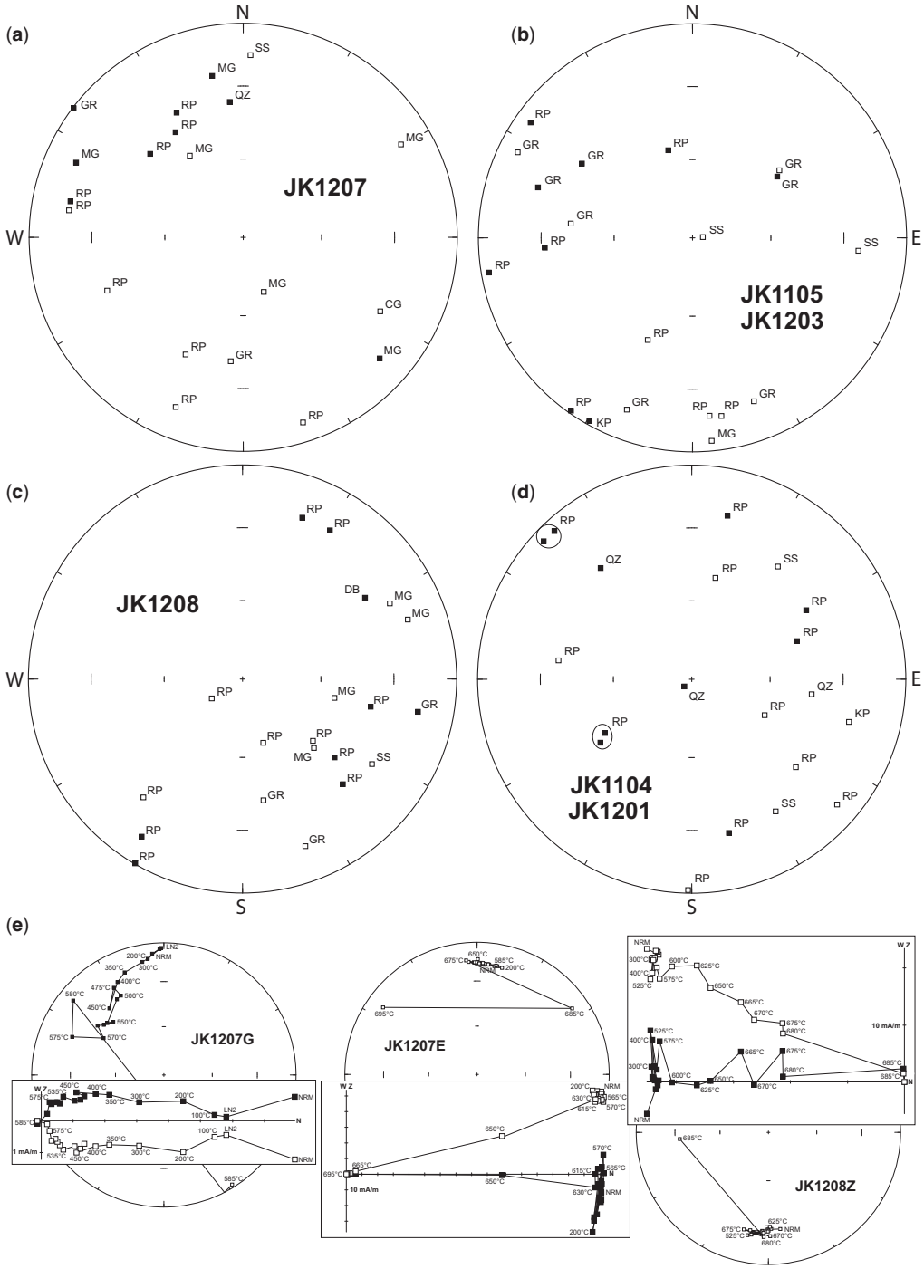


Fig. 6. Results of conglomerate tests in the Aubures Formation. In (A)–(D), equal-area sterographs follow the same symbol convention as in Figure 5. In (E), representative specimen demagnetization behaviours follow the same conventions as in Figure 4. Encircled pairs of data indicate double-sampled clasts for consistency tests. Abbreviations for clast lithologies: CG, onglomerate; GR, granitoid (medium-coarse grained); KP, K-feldspar porphyry; MG, microgranite/granophyre; QZ, quartzite; RP, rhyolite porphyry; SS, sandstone.

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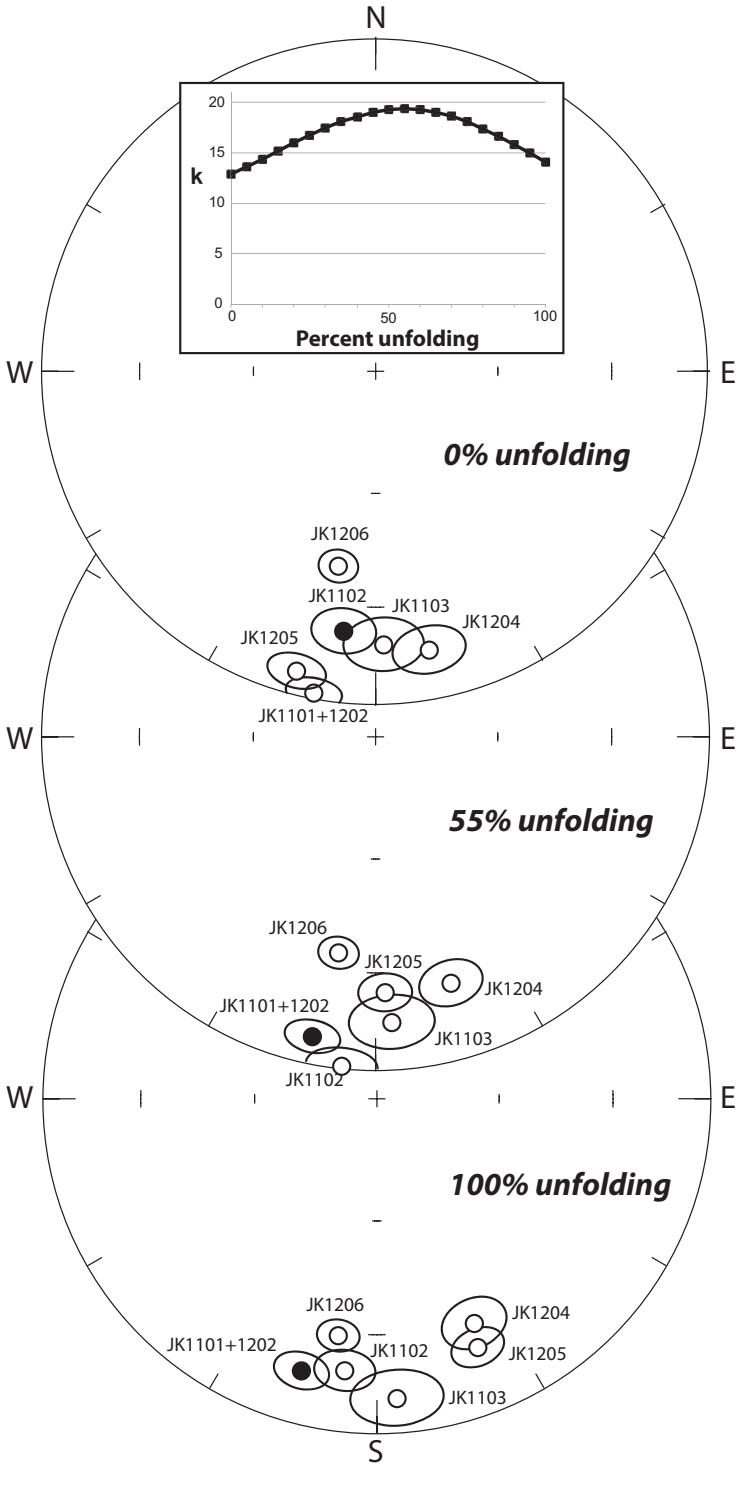


Fig. 7. Results of the regional fold test. Inset: Fisher's (1953) precision parameter for the mean of six locality means, according to sequential proportions of unfolding. Maximum clustering of data occurs at 55% unfolding.

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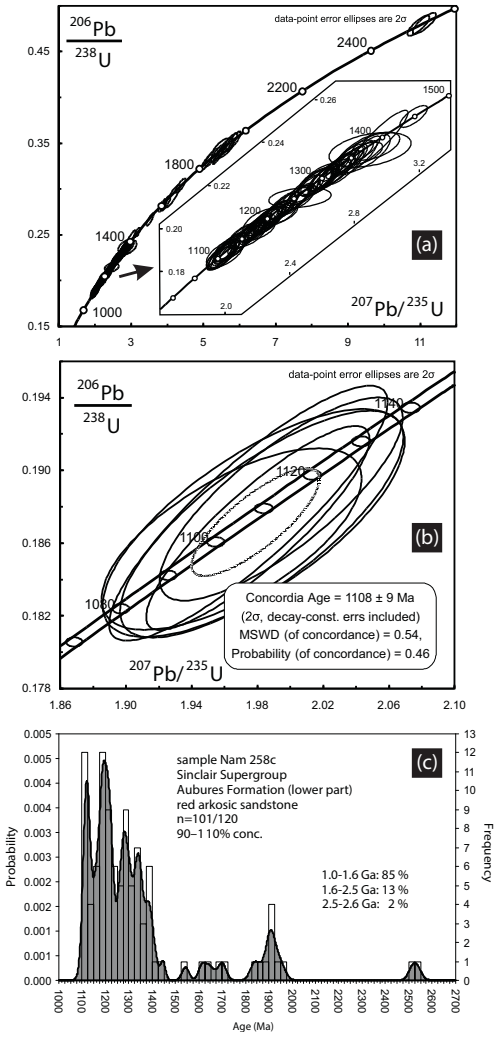


Fig. 8. LA-ICP-MS U–Pb ages of detrital zircon grains from the Aubures Formation. (**a**) Concordia diagram of all 90–110% concordant grains (error ellipses based on a 2σ error level). (**b**) U–Pb concordia age of the youngest population of six detrital zircon grains providing a maximum age of sedimentation of 1108 $\pm$ 9 Ma. (**c**) Combined binned frequency and probability density distribution plots (AgeDisplay) of all concordant analyses, based on a 1σ error level. Bin widths are 25 Ma.

1.7–1.5 Ga range, but notably, the prominent late Palaeoproterozoic (1.8–1.7 Ga) granitic magmatism in the Rehoboth basement inlier (Van Schijndel *et al.* 2014) is not represented in Aubures detritus.

Within the youngest, most prominent mode of zircon ages from the 1.4–1.1 Ga interval, submodes appear separated by about 100 myr intervals, reminiscent of the distinct magmatic pulses within the Sinclair region itself (Hoal & Heaman 1995). The youngest such pulse, represented by the six youngest analyses (a10, b9, b26, b40, b44, and b60; Table 2), yields a concordia U–Pb age of 1108 $\pm$ 9 Ma (Fig. 8b).

Discussion

Correlation of redbeds across the western Kalahari craton

Previous syntheses of the western Kalahari Mesoproterozoic geological record have emphasized the similarity of post-tectonic redbed successions from central Namibia to the Upington region of South Africa (e.g. SACS 1980; Becker *et al.* 2006; Miller 2012). The youngest population of U–Pb detrital zircon ages found in this study, with a mean of 1108 $\pm$ 9 Ma, constitutes a maximum age constraint for deposition of the Aubures Formation. In order to provide additional age constraints, the new Aubures palaeomagnetic pole is compared with the Kalahari craton's fragmental apparent polar wander (APW) path (Fig. 9, Table 3). The Aubures pole is located between poles of the Umkondo large igneous province (Gose *et al.* 2006), dated at 1112 $\pm$ 0.5 Ma to 1108 $\pm$ 0.9 Ma (Hanson *et al.* 2004) and that of the Kalkpunt Formation from the Koras Group, located near Upington, South Africa (Briden *et al.* 1979). The Kalkpunt redbeds have a maximum and probably approximate age of 1093 $\pm$ 7 Ma derived from a conformably underlying rhyolite flow (Pettersson *et al.* 2007). The Kalahari APW path migrates rapidly away from these poles towards a distant southwestern position at *c.* 1000 Ma (Gose *et al.* 2006), thus our new Aubures palaeomagnetic result suggests an age of *c.* 1100 Ma for its deposition. The new pole also supports earlier conclusions that the Sinclair succession was attached to the western margin of the Kalahari craton by at least 1100 Ma, if not substantially earlier (Becker *et al.* 2006; Miller 2012).

Proximity of our new Aubures palaeomagnetic pole to that of the Kalkpunt Formation also provides further evidence for correlation between the Konkiep Group in central Namibia, with the Koras Group in South Africa, *c.* 800 km SE (SACS 1980; Fig. 1). These successions are similar in their general profile of a deformed, metamorphosed basement being overlain by a bimodal suite of relatively undeformed volcanic rocks (Miller 2008; Bailie *et al.* 2012). Both groups contain siliciclastic metasediments, at no more than subgreenschist grade within and overlying the volcanic suite (Pettersson *et al.* 2007; Miller 2008). The groups are both found in fault-bounded basins that appear to be related to an upper-plate subduction setting

Table 2. *LA-SF-ICP-MS U–Pb–Th data of detrital zircon from sample Nam 258c, n = 102 of 120 analysed zircon grains; red arkosic sandstone, Aubures Formation (lower part), Sinclair Supergroup, Mesoproterozoic, roadcut NW of Helmeringhausen, Namibia, coordinates: 25°44′24.70″S; 16°24′53.90″S*

Number	$^{207}Pb^{*}$ (cps)	(ppm)	$Pb^{\dagger}$ (ppm)	$\frac{Th^{\dagger}}{U}$	$\frac{^{206}Pb^{\ddagger}}{^{204}Pb}$	$\frac{^{206}Pb^{\ddagger}}{^{238}U}$	2σ (%)	$\frac{^{207}Pb^{\ddagger}}{^{235}U}$	2σ (%)	$\frac{^{207}Pb^{\ddagger}}{^{206}Pb}$	2σ (%)	ρ§	$\frac{^{206}Pb}{^{238}U}$	2σ (Ma)	$\frac{^{207}Pb}{^{235}U}$	2σ (Ma)	$\frac{^{207}Pb}{^{206}Pb}$	2σ (Ma)	Concentration (%)
b60	5154	11	3	2.90	5408	0.18622	2.2	1.96416	3.3	0.07650	2.5	0.66	1101	22	1103	22	1108	49	99
b40	49235	223	40	0.09	14867	0.18683	2.5	1.98019	2.8	0.07687	1.4	0.87	1104	25	1109	19	1118	27	99
b9	8062	33	7	0.96	4637	0.18704	2.7	1.98256	3.6	0.07688	2.3	0.77	1105	28	1110	24	1118	45	99
b44	58310	139	31	1.05	356	0.18710	2.7	1.97772	3.8	0.07666	2.6	0.72	1106	28	1108	26	1112	53	99
b48	96998	226	46	0.31	720	0.18730	3.4	2.65574	4.8	0.10284	3.4	0.71	1107	35	1316	36	1676	63	66
b26	141262	607	112	0.11	1329	0.18764	3.1	1.97445	3.5	0.07632	1.7	0.87	1109	31	1107	24	1103	35	100
a10	19754	63	14	1.04	25791	0.18806	1.8	1.99751	2.4	0.07704	1.6	0.75	1111	19	1115	17	1122	32	99
a12	6425	25	6	1.05	5482	0.18827	2.9	1.99350	3.8	0.07679	2.5	0.76	1112	30	1113	26	1116	49	100
a23	3179	12	3	1.35	2612	0.18835	2.8	1.99864	4.6	0.07696	3.7	0.60	1112	29	1115	32	1120	74	99
b13	20878	88	27	2.73	15453	0.18842	2.4	1.99446	3.0	0.07677	1.8	0.80	1113	25	1114	21	1115	36	100
a9	23500	76	18	1.33	8083	0.18910	1.9	1.99816	2.8	0.07664	2.0	0.68	1116	20	1115	19	1112	41	100
b46	44413	133	24	0.09	57706	0.18944	2.2	2.02907	2.6	0.07768	1.4	0.84	1118	22	1125	18	1139	27	98
a3	120630	300	63	0.57	518	0.18952	2.6	2.01263	3.2	0.07702	1.9	0.82	1119	27	1120	22	1122	37	100
a24	4311	16	5	2.70	5624	0.19111	2.3	2.03389	3.9	0.07718	3.1	0.60	1127	24	1127	27	1126	62	100
a60	20497	32	6	0.33	8748	0.19141	2.8	2.04515	3.9	0.07749	2.7	0.72	1129	29	1131	27	1134	54	100
a47	18379	49	9	0.27	4632	0.19151	2.4	2.03692	2.9	0.07714	1.6	0.83	1130	25	1128	20	1125	33	100
b42	25746	93	21	1.02	1344	0.19171	2.8	2.04522	3.9	0.07738	2.7	0.71	1131	29	1131	27	1131	55	100
b34	17871	87	18	0.48	5180	0.19456	2.8	2.10298	3.3	0.07839	1.8	0.85	1146	29	1150	23	1157	35	99
b43	11978	50	10	0.58	15483	0.19575	2.2	2.11433	3.1	0.07834	2.1	0.72	1152	24	1153	22	1155	42	100
b23	11076	46	10	0.67	3610	0.19621	2.4	2.11588	3.6	0.07821	2.6	0.67	1155	25	1154	25	1152	53	100
a14	13514	48	10	0.49	15984	0.19617	2.1	2.13114	2.8	0.07879	1.9	0.75	1155	23	1159	20	1167	37	99
b39	7013	29	6	0.57	7472	0.19697	2.4	2.13816	4.2	0.07873	3.4	0.57	1159	26	1161	29	1165	68	99
b24	12423	60	13	0.77	11224	0.19715	2.9	2.13897	3.2	0.07869	1.5	0.88	1160	30	1161	23	1164	30	100
a46	39827	99	19	0.08	5343	0.19845	2.0	2.17822	2.3	0.07961	1.1	0.87	1167	22	1174	16	1187	22	98
a48	29392	63	13	0.54	12649	0.19845	1.8	2.17827	2.5	0.07961	1.7	0.74	1167	20	1174	17	1187	33	98
a53	34792	70	15	0.36	3271	0.19863	2.4	2.16828	3.1	0.07917	1.9	0.79	1168	26	1171	22	1176	38	99
b10	16369	60	12	0.48	7119	0.19872	2.1	2.18002	3.1	0.07956	2.3	0.69	1168	23	1175	22	1186	45	99
a31	10176	35	9	1.59	1476	0.19980	2.1	2.20915	3.8	0.08019	3.2	0.55	1174	22	1184	27	1202	62	98
b53	42995	101	20	0.21	2487	0.20005	2.5	2.18471	2.9	0.07921	1.4	0.87	1176	27	1176	20	1177	29	100
a54	29197	55	12	0.53	31926	0.20034	2.3	2.18557	3.0	0.07912	1.9	0.78	1177	25	1176	21	1175	37	100
b30	17304	78	17	0.83	10347	0.20088	2.2	2.20280	2.6	0.07953	1.4	0.83	1180	23	1182	18	1185	28	100
b16	32812	127	26	0.33	41559	0.20134	2.4	2.21215	2.9	0.07969	1.6	0.82	1183	26	1185	20	1189	33	99
b28	33201	138	29	0.41	14289	0.20144	2.2	2.21787	2.6	0.07985	1.3	0.87	1183	24	1187	18	1193	25	99
a41	32773	85	17	0.25	40920	0.20184	1.8	2.24463	2.2	0.08066	1.3	0.82	1185	20	1195	16	1213	25	98
b38	3847	15	4	0.91	4855	0.20306	2.5	2.22757	4.3	0.07956	3.5	0.58	1192	27	1190	31	1186	69	100

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a58	39122	50	11	0.51	1034	0.20355	2.6	2.25617	3.6	0.08039	2.5	0.72	1194	29	1199	26	1207	50	99
a25	16588	58	13	0.72	2713	0.20343	2.7	2.25870	3.4	0.08053	2.0	0.80	1194	30	1200	24	1210	40	99
a2	7931	22	5	0.53	2652	0.20374	2.1	2.24819	3.0	0.08003	2.1	0.70	1195	23	1196	21	1198	42	100
a34	16291	60	13	0.60	5787	0.20392	2.7	2.26496	3.2	0.08056	1.8	0.82	1196	29	1201	23	1211	36	99
b54	10076	25	5	0.40	12707	0.20422	2.2	2.25423	2.7	0.08006	1.6	0.82	1198	24	1198	19	1198	31	100
b18	45267	162	34	0.33	56778	0.20630	2.4	2.28941	2.8	0.08049	1.4	0.86	1209	27	1209	20	1209	28	100
a1	47265	139	27	0.07	59060	0.20633	2.0	2.29568	2.9	0.08069	2.1	0.69	1209	22	1211	20	1214	41	100
a8	23000	67	19	1.89	20169	0.20672	2.2	2.29406	2.7	0.08048	1.5	0.82	1211	24	1210	19	1209	30	100
a16	12219	42	11	1.55	1420	0.20722	1.7	2.35841	3.1	0.08254	2.6	0.56	1214	19	1230	22	1258	50	96
a39	5490	14	3	0.57	3341	0.20732	2.3	2.34199	3.6	0.08193	2.8	0.64	1215	26	1225	26	1244	55	98
a51	26315	52	15	1.54	7330	0.20804	2.0	2.32833	2.7	0.08117	1.9	0.74	1218	22	1221	20	1226	36	99
a52	91521	158	33	0.21	1618	0.20811	3.2	2.32863	3.8	0.08115	2.0	0.85	1219	36	1221	27	1225	39	99
a30	30019	101	22	0.51	1251	0.20860	1.8	2.33824	3.0	0.08129	2.5	0.58	1221	20	1224	22	1229	48	99
b50	30501	78	16	0.25	4848	0.20966	2.9	2.34825	3.7	0.08123	2.2	0.80	1227	33	1227	26	1227	43	100
b8	29649	97	22	0.64	36878	0.21060	2.3	2.35862	2.8	0.08123	1.7	0.80	1232	25	1230	20	1227	33	100
a15	50366	133	29	0.37	2773	0.21248	2.2	2.44198	2.6	0.08335	1.3	0.86	1242	25	1255	19	1277	25	97
a40	21742	54	12	0.46	4657	0.21367	2.4	2.42573	3.1	0.08234	2.0	0.77	1248	27	1250	23	1254	40	100
b11	24638	70	17	0.75	29991	0.21389	2.5	2.44428	3.0	0.08288	1.7	0.82	1250	28	1256	22	1266	33	99
a13	63126	171	48	1.50	19963	0.21438	1.8	2.46265	6.4	0.08331	6.1	0.28	1252	20	1261	47	1277	120	98
b57	40904	87	20	0.57	2283	0.21453	2.0	2.49162	2.4	0.08423	1.3	0.83	1253	23	1270	17	1298	26	97
a26	143124	429	95	0.32	1844	0.21466	2.5	2.45347	3.0	0.08290	1.7	0.83	1254	28	1258	22	1267	32	99
a59	41189	63	15	0.65	23144	0.21583	1.8	2.48120	2.7	0.08338	2.0	0.66	1260	20	1267	19	1278	39	99
b14	19701	58	16	1.39	23925	0.21778	2.7	2.48873	3.4	0.08288	2.0	0.80	1270	31	1269	25	1266	39	100
a42	18752	47	15	2.03	6178	0.21873	2.3	2.51738	3.0	0.08347	1.9	0.77	1275	27	1277	22	1280	38	100
b17	20478	69	15	0.27	5567	0.22144	2.3	2.56184	3.5	0.08391	2.6	0.67	1289	27	1290	26	1290	50	100
a35	22638	71	16	0.32	1232	0.22192	2.4	2.56856	3.7	0.08395	2.8	0.65	1292	29	1292	28	1291	55	100
b59	8583	16	4	1.25	5793	0.22196	2.2	2.56822	3.2	0.08392	2.2	0.71	1292	26	1292	23	1291	44	100
b1	51231	129	29	0.16	2418	0.22273	2.8	2.58229	4.0	0.08409	2.8	0.70	1296	33	1296	29	1295	55	100
b55	37361	72	17	0.57	9416	0.22343	2.4	2.60077	2.9	0.08442	1.6	0.83	1300	28	1301	21	1302	31	100
a11	20700	53	14	0.94	17257	0.22358	2.3	2.61054	2.8	0.08468	1.7	0.80	1301	27	1304	21	1308	33	99
b12	46239	149	34	0.34	5606	0.22358	2.0	2.62440	2.5	0.08513	1.4	0.82	1301	24	1307	18	1319	27	99
a29	22541	68	16	0.34	26573	0.22425	2.0	2.65608	2.7	0.08590	1.7	0.76	1304	24	1316	20	1336	33	98
a21	16675	45	11	0.47	6667	0.22492	2.6	2.62243	3.0	0.08456	1.5	0.87	1308	31	1307	22	1306	28	100
a36	26162	73	18	0.51	30718	0.22620	2.2	2.67810	3.1	0.08587	2.1	0.73	1315	27	1322	23	1335	40	98
b4	14350	41	10	0.57	16851	0.22718	2.7	2.69526	3.3	0.08605	2.0	0.80	1320	32	1327	25	1339	39	99
b22	25178	81	19	0.46	29741	0.22856	2.3	2.68541	2.7	0.08521	1.4	0.85	1327	28	1324	20	1320	27	100
a32	11778	41	10	0.64	13607	0.22848	2.8	2.74973	3.2	0.08728	1.6	0.86	1327	33	1342	24	1367	31	97
b33	28983	104	26	0.66	34115	0.22919	2.1	2.70982	2.6	0.08575	1.5	0.81	1330	26	1331	20	1333	30	100
a43	16852	40	10	0.75	1460	0.23111	1.9	2.74376	2.5	0.08611	1.6	0.77	1340	24	1340	19	1341	31	100
b41	18144	52	13	0.74	21175	0.23103	2.5	2.75475	2.9	0.08648	1.5	0.85	1340	30	1343	22	1349	30	99
a49	22533	39	11	0.91	4128	0.23226	2.5	2.76562	3.3	0.08636	2.2	0.76	1346	31	1346	25	1346	42	100

(*Continued*)

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Table 2. *LA-SF-ICP-MS U–Pb–Th data of detrital zircon from sample Nam 258c, n = 102 of 120 analysed zircon grains; red arkosic sandstone, Aubures Formation (lower part), Sinclair Supergroup, Mesoproterozoic, roadcut NW of Helmeringhausen, Namibia, coordinates: 25°44′24.70″S; 16°24′53.90″S (Continued)*

Number	^{207}Pb* (cps)	(ppm)	$Pb^{\dagger}$ (ppm)	$\frac{Th^{\dagger}}{U}$	$\frac{^{206}Pb^{\ddagger}}{^{204}Pb}$	$\frac{^{206}Pb^{\ddagger}}{^{238}U}$	2σ (%)	$\frac{^{207}Pb^{\ddagger}}{^{235}U}$	2σ (%)	$\frac{^{207}Pb^{\ddagger}}{^{206}Pb}$	2σ (%)	ρ§	$\frac{^{206}Pb}{^{238}U}$	2σ (Ma)	$\frac{^{207}Pb}{^{235}U}$	2σ (Ma)	$\frac{^{207}Pb}{^{206}Pb}$	2σ (Ma)	Concentration (%)
b47	38797	84	21	0.62	21960	0.23231	2.2	2.77374	2.6	0.08660	1.4	0.85	1347	27	1348	20	1352	27	100
b15	33415	100	23	0.26	11043	0.23302	2.8	2.82504	3.3	0.08793	1.8	0.84	1350	34	1362	25	1381	34	98
b32	8658	32	8	0.48	2529	0.23511	2.2	2.81146	3.0	0.08673	2.1	0.71	1361	27	1359	23	1354	41	100
a18	36103	102	25	0.53	1389	0.23516	1.8	2.86385	3.1	0.08832	2.5	0.58	1361	22	1372	24	1390	49	98
b7	16024	37	12	1.48	908	0.23522	3.2	2.85367	4.7	0.08799	3.5	0.67	1362	39	1370	36	1382	68	99
a38	15408	43	12	1.01	682	0.23668	2.8	2.89564	7.1	0.08873	6.6	0.39	1369	34	1381	55	1398	126	98
a17	62744	137	35	0.48	1011	0.23751	1.9	2.87484	2.4	0.08779	1.4	0.81	1374	24	1375	18	1378	27	100
b27	14855	51	14	0.61	2436	0.23839	2.2	2.92944	4.2	0.08912	3.5	0.53	1378	28	1390	32	1407	67	98
b37	29061	100	27	0.78	6570	0.24165	4.5	2.94853	5.0	0.08849	2.1	0.91	1395	57	1394	39	1393	40	100
a4	119896	230	65	0.71	2609	0.25225	1.6	3.16448	2.2	0.09098	1.5	0.75	1450	21	1449	17	1446	28	100
b5	128343	265	74	0.38	2494	0.26677	2.2	3.52257	2.5	0.09577	1.3	0.86	1524	29	1532	20	1543	24	99
a28	29826	64	20	0.67	2266	0.28201	2.0	3.86348	2.5	0.09936	1.4	0.82	1601	29	1606	20	1612	26	99
a37	34261	65	22	0.85	6315	0.28429	2.3	3.95418	2.8	0.10088	1.6	0.82	1613	33	1625	23	1640	29	98
b25	94403	268	81	0.31	1386	0.29209	3.5	4.14727	4.0	0.10298	1.9	0.87	1652	51	1664	33	1679	36	98
a33	36877	69	23	0.67	29992	0.30363	2.1	4.36796	2.6	0.10434	1.5	0.82	1709	32	1706	21	1703	27	100
a50	116016	111	38	0.25	15537	0.32909	2.1	5.08454	2.5	0.11206	1.2	0.87	1834	34	1834	21	1833	22	100
b3	35785	52	20	0.73	21277	0.33413	2.3	5.22148	2.9	0.11334	1.6	0.82	1858	38	1856	25	1854	29	100
b20	97314	156	55	0.30	1864	0.33636	2.2	5.42429	2.4	0.11696	0.8	0.94	1869	36	1889	21	1910	14	98
b45	42016	53	22	1.13	36718	0.33868	2.2	5.38879	2.7	0.11540	1.4	0.84	1880	37	1883	23	1886	26	100
a6	84739	97	38	0.75	26711	0.34238	1.7	5.50694	1.9	0.11666	0.8	0.90	1898	29	1902	17	1906	15	100
a5	24814	32	12	0.63	2618	0.34427	2.7	5.53781	4.3	0.11666	3.4	0.62	1907	44	1906	37	1906	60	100
b51	50216	52	21	0.97	4863	0.34620	2.3	5.61038	2.8	0.11753	1.6	0.83	1916	39	1918	25	1919	28	100
b6	94850	131	49	0.41	9331	0.35063	2.4	5.73667	2.8	0.11866	1.5	0.84	1938	40	1937	25	1936	28	100
b19	45008	60	26	1.11	37798	0.35215	2.4	5.82523	2.9	0.11997	1.5	0.84	1945	41	1950	25	1956	28	99
a19	68634	38	23	1.02	41664	0.47824	1.9	10.94735	2.3	0.16602	1.3	0.83	2520	40	2519	22	2518	22	100
a20	52408	32	20	1.21	31496	0.48073	1.9	11.11371	2.4	0.16767	1.4	0.81	2530	41	2533	23	2535	23	100

*Within-run background-corrected mean ^{207}Pb signal in counts per second.

†U and Pb content and Th/U ratio were calculated relative to GJ-1 and are accurate to approximately 10%.

‡corrected for background, mass bias, laser induced U–Pb fractionation and common Pb (if detectable, see analytical method) using Stacey & Kramers (1975) model Pb composition. $^{207}Pb/^{235}U$ calculated using $^{207}Pb/^{206}Pb/(^{238}U/^{206}Pb \times 1/137.88)$. Errors are propagated by quadratic addition of within-run errors (2SE) and the reproducibility of GJ-1 (2SD).

§ρ is the error correlation defined as $err^{206}Pb/^{238}U/err^{207}Pb/^{235}$.

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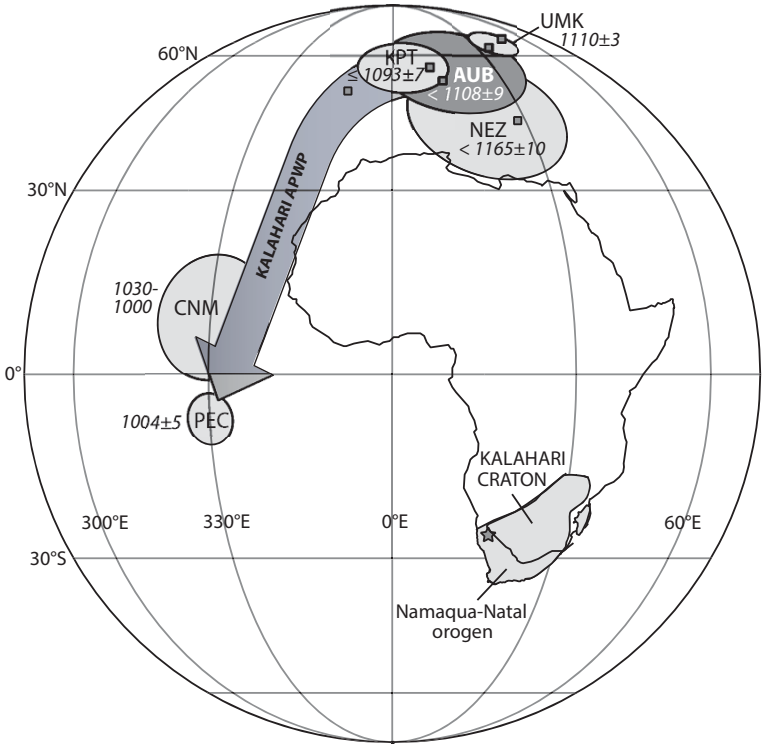


Fig. 9. Apparent polar wander (APW) path for the Kalahari craton between *c.* 1200 and 1000 Ma (orthographic projection). Pole codes are explained in Table 3. Small squares are the virtual geomagnetic poles (VGPs) from the six Aubures Formation locality-mean directions, each upon 55% unfolding. Ages in million years. Star, Sinclair sampling region.

characterized by alternating phases of compression and extension. Like the Aubures Formation, the Kalkpunt Formation is a red sandstone unit overlying a bimodal volcanic succession, containing a basal conglomerate as well as conglomerate lenses throughout its stratigraphy, and shale near the top of the formation. The Kalkpunt depositional environment has been interpreted as an alluvial fan developing in a synclinal structure along the fault scarps bounding the basin (SACS 1980); facies distribution was probably controlled by basement highs and syndepositional faulting, following an early Namaqua orogenic event at *c.* 1200 Ma (Gutzmer *et al.* 2000; Pettersson *et al.* 2007; Bailie *et al.* 2012). Finally, the detrital zircon age spectra from Aubures Formation, as documented in this study, are similar to the few existing U–Pb detrital zircon analyses of the Kalkpunt Formation (Pettersson *et al.* 2007). Whether or not the Konkiep and Koras basins were originally closer together, and subsequently separated by dextral motion along various NW-striking shear zones of the Namaqua Province (Stowe 1983; Miller 2012), is a question that is not resolvable by the presently available palaeomagnetic data. The Aubures and Kalkpunt poles are compatible with zero tectonic offset along Namaqua shear zones, but the 95% confidence radii of both poles would also permit relative displacement on the order of several hundred kilometers.

In the Rehoboth basement inlier, post-tectonic redbeds of the Tsumis Group include the Eskadron, Doornpoort and Klein Aub Formations, the latter hosting a significant copper deposit (Becker & Schalk 2008). They rest unconformably atop all other Rehoboth basement units, including the Langberg Formation that yielded detrital ages as young as 1103 $\pm$ 24 Ma (Van Schijndel *et al.* 2011). In fact, the Langberg and Aubures detrital zircon age populations are remarkably similar, although the Langberg strata have generally been correlated with Guperas strata based on a substantial volcanogenic component and a tectonic fabric greater than that experienced by the Aubures (Becker & Schalk 2008). Strong deformation and greenschist-facies metamorphism of the Langberg Formation at the U–Pb sampled locality (Van Schijndel *et al.* 2011) imply a post-1100 Ma tectonic event in the Rehoboth region, followed by still younger redbed

Table 3. *Kalahari and Sinclair palaeomagnetic poles shown in Figure 9*

Rock name	Code	Age (Ma)	Latitude (°N)	Longitude (°E)	A_{95} (deg)	1	2	3	4	5	6	7	Q	Reference(s)
Namaqua Eastern Zone	NEZ	$<1165 \pm 10$	45	022	13	0	1	1	1	0	0	0	3	Evans *et al.* 2002; Pettersson *et al.* 2007
Umkondo grand mean	UMK	1110 ± 3	64	039	4	1	1	1	1	1	1	1	7	Gose *et al.* 2006
Aubures Formation	AUB	$<1108 \pm 9$	56	018	11	0	1	1	1	1	1	0	5	*this study*
Kalkpunt Formation	KPT	$\leq 1093 \pm 7$	57	003	7	1	0	1	0	1	0	0	3	Briden *et al.* 1979; Pettersson *et al.* 2007
Central Namaqua belt	CNM	1030–1000	08	330	10	1	1	1	0	0	1	0	4	Onstott *et al.* 1986; Powell *et al.* 2001
Port Edward Charnockite	PEC	1004 ± 5	−07	330	4	1	1	1	0	0	0	0	3	Gose *et al.* 2004

Note: Q is the reliability factor of Van der Voo (1990).

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sedimentation of the Tsumis Group. Because our Aubures palaeomagnetic pole is well constrained on the Kalahari APW path at *c.* 1100 Ma, it would appear that the Tsumis Group redbeds are younger than the Aubures and Kalkpunt Formations. Previous palaeomagnetic study of the Doornpoort redbeds, unfortunately, only identified a post-folding overprint (Piper 1975).

Implications for palaeogeographic reconstructions

Because our Aubures palaeomagnetic data are comparable to existing high-quality poles from Kalahari at *c.* 1100 Ma, we add little new palaeolatitude information for use in global reconstructions during the assembly phase of the Rodinia supercontinent (cf. Li *et al.* 2008). However, the dominant magnetic polarity of our Aubures data is directed toward the south, and this may be useful for constraining orientation options for the low-latitude Kalahari craton at that time (cf. Hanson *et al.* 2004). The Aubures polarity dominance is the same as that from the Umkondo large igneous province, despite the fact that our Aubures detrital zircon data indicate deposition after Umkondo emplacement. Hanson *et al.* (2004) interpreted the south-directed, dominant Umkondo remanence as reversed polarity, but such an interpretation would imply *c.* 180° relative rotation between Kalahari

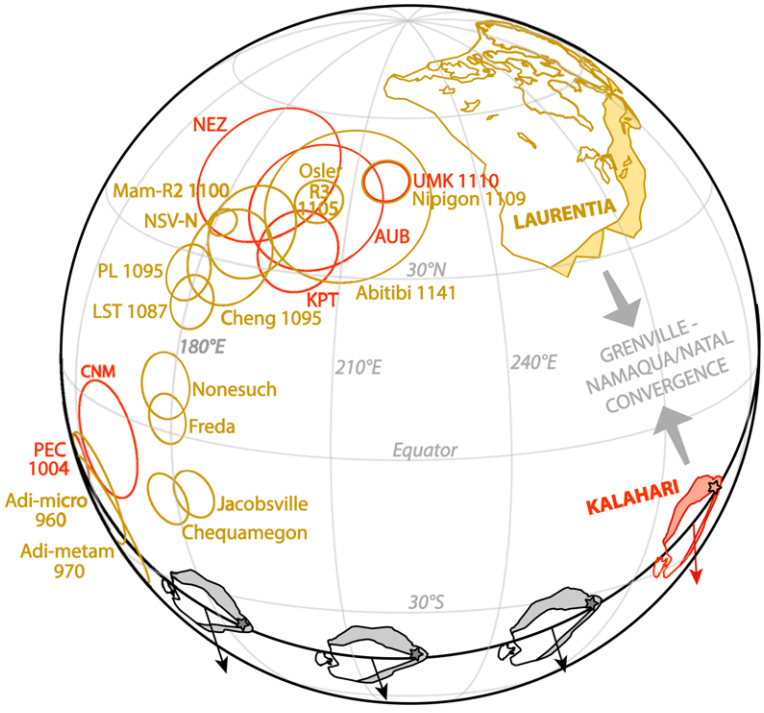


Fig. 10. Relative reconstruction of Kalahari and Laurentia cratons at 1110 Ma, in the present North American reference frame and orthographic projection, assuming that the dominant polarity of Aubures Formation remanence is geomagnetically Normal. The highest-quality Kalahari pole, Umkondo (Gose *et al.* 2006), is placed directly atop the coeval pole from Laurentia, Nipigon intrusions (Pisarevsky *et al.* 2014). Palaeolongitude freedom around that pole pairing allows numerous alternative locations of Kalahari (greyscale maps; present north indicated by each arrow; stars show the Sinclair region) along the 1110 Ma palaeoequator (black curve). Red Kalahari map and corresponding poles (Table 3) are restored to Laurentia (04.1°, 334.7°, −117.1°) to align, additionally, the Aubures and Kalkpunt poles with those from Laurentia at *c.* 1100 Ma, conforming to a true polar wander interpretation for the 1110–1090 Ma APW path (Swanson-Hysell *et al.* 2009). Laurentia poles: Abitibi dykes and Nipigon means (Pisarevsky *et al.* 2014), Osler R3, upper third of the Osler reversed volcanic rocks (Swanson-Hysell *et al.* 2014*a*); Mam-R2, Mamainse Point upper reversed flows (Swanson-Hysell *et al.* 2009, 2014*b*); NSV-N, North Shore Volcanics normal (Tauxe & Kodama 2009); Cheng, Chengwatana volcanics (Kean *et al.* 1997; Zartman *et al.* 1997); PL, Portage Lake volcanics (Hnat *et al.* 2006); LST, Lake Shore traps (Diehl & Haig 1994); Nonesuch (Symons *et al.* 2013), Freda (Henry *et al.* 1977), Jacobsville (Roy & Robertson 1978), Chequamegon (McCabe & Van der Voo 1983). Adi-metam, Adirondack metamorphosed anorthosites (Brown & McEnroe 2012); Adi-micro, Adirondack microcline gneisses (Brown & McEnroe 2012).

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and Laurentia if the two are considered to have collided along a joint Grenville(Llano)/Namaqua–Natal suture (Dalziel *et al.* 2000; Jacobs *et al.* 2008; Loewy *et al.* 2011). As Jacobs *et al.* (2008) noted, however, the Reversed polarity interpretation for the majority of Umkondo data is merely suggestive, not definitive. The near entirety of Aubures Formation, spanning at least 1–2 km of fluvial-deltaic stratigraphy, has a south-directed palaeomagnetic remanence. Given that the Aubures contains detrital zircons as young as 1108 ± 9 Ma, and that it may correlate with the Kalkpunt Formation that is probably only slightly younger than 1093 ± 7 Ma, we suspect that an age of *c.* 1100–1090 Ma for the majority of Aubures sedimentation probably occurred during the Normal polarity (super?)chron coincident with younger Keweenawan volcanism on Laurentia.

Inverting the absolute polarity interpretation of *c.* 1100 Ma Kalahari palaeomagnetic poles carries important tectonic and palaeogeographic implications. First, it permits the Grenville (Llano) and Namaqua–Natal belts to face each other across a diminishing ocean, perhaps to collide head-on in the succeeding tens of millions of years to help build the Rodinia supercontinent (cf. Loewy *et al.* 2011). Figure 10 shows a precise superposition of the high-quality Umkondo and Nipigon intrusions poles, from Kalahari and Laurentia, respectively. By aligning just those poles, there would be freedom of palaeolongitude in Kalahari reconstructions relative to Laurentia, around a small circle about the joint palaeopole position. Four possible reconstructions of Kalahari are shown in the figure, all straddling the Umkondo palaeoequator (black curve). Second, because Laurentia and Kalahari should have lain on separate, converging plates, any similarity in the lengths and shapes of their APW paths, if not entirely fortuitous, would need to be attributed to true polar wander (TPW), which is the uniform motion of the solid Earth relative to its rotation axis. To our knowledge, Powell *et al.* (2001) were the first to suggest a TPW interpretation for the rapid continental motion implied by *c.* 1100 Ma poles from Kalahari and Laurentia, an idea thereafter incorporated into a general supercontinent-geodynamic model by Evans (2003) and quantified by Swanson-Hysell *et al.* (2009, 2014*a*) to a rate of *c.* 24 cm per year for Laurentia's palaeolatitude changes. Such rapid rates of TPW would probably dwarf contemporaneous rates of relative plate motions, thus justifying a superposition of Aubures and Kalkpunt poles atop *c.* 1100 Ma Keweenawan poles and generating our preferred (red colour) reconstruction shown in Figure 10. Agreement of the *c.* 1000 Ma poles from both cratons in the same reconstruction, however, should be coincidental, given the likely substantial convergence between Laurentia and Kalahari between 1100 and 1000 Ma (e.g. Loewy *et al.* 2011). Such a coincidence arises, according to our palaeogeographic model, because the 1000 Ma palaeomagnetic poles would have lain near the Euler stage pole of relative motion between the converging plates through that 100 myr interval.

Conclusions

In this study, we present a new palaeomagnetic pole and new detrital zircon U–Pb data for the Mesoproterozoic Aubures Formation. U–Pb ages of zircon grains result in a maximum age of sedimentation of *c.* 1108 Ma for the Aubures Formation. The main age group of 1.4–1.1 Ga and can be explained by erosion of the Namaqua orogen along the western margin of the Kalahari craton, as well as input from the Umkondo large igneous province. Bolstered by detailed thermal demagnetization laboratory techniques, a positive intraformational conglomerate field stability test and maximal clustering of site-mean data at 55% untilting, our palaeomagnetic data demonstrate a primary magnetization that was probably contemporaneous with syndepositional folding during rift basin development. The pole overlaps with that of the Kalkpunt Formation at the top of the Koras Group in South Africa, suggesting a temporal correlation between the two redbed formations. Redbed sedimentation was widespread along the western Kalahari craton margin after the first phase of Namaqua belt orogenesis (*c.* 1200 Ma) and Umkondo large igneous province emplacement (*c.* 1110 Ma). Palaeomagnetic polarity patterns between groups of poles from Kalahari and Laurentia from *c.* 1100 Ma can help constrain global palaeogeography during Rodinia supercontinent assembly.

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